

COMP SCI/SFWR ENG 4/6E03 – Performance Analysis of Computer Systems

Instructor: Douglas Down

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Office Hours: (instructor) Monday, Thursday 1:30-3:30. I am also available by appointment (if you prefer to meet online, for example).

Lectures: Monday, Wednesday, Thursday 10:30 (ABB/102)

Tutorials: T01 – Thursday 12:30 (ITB/139); T02 – Thursday 11:30 (BSB/104); T03 - Wednesday 1:30 (BSB/120).

TA contact: Zhaleh Rahimi (rahimz1@mcmaster.ca), Maryam Akbari Moghaddam (akbarimm@mcmaster.ca), Nafiseh Valizadeh Shiran (valizan@mcmaster.ca).

Mission: The mission of this course is to teach students how to use mathematical modelling and simulation to estimate quantitative aspects of a computer system's behaviour. They should also learn how modelling results can guide them in the selection of a computer system's hardware and the design of the software. The course should provide students with experience in studying a proposed design and producing a model, deciding whether there are explicit symbolic solutions to the problem or whether simulation should be used. Students should be able to evaluate the effect of using simplified models so that solutions can be found.

Grading:

Two term tests	20% each
Assignments	15%
Final exam (2.5 hours)	45%

There will be eleven assignments (one per week), your assignment mark will consist of the best six of these. There will be no extensions for assignment due dates.

For those taking the course at the 600-level, there will be an extra project to be mutually agreed upon. This will be done on an individual basis. The marking scheme for graduate students will be 15% for each of the term tests, 10% for the assignments, 40% for the final exam and 20% for the project.

(The instructor reserves the right to conduct deferred examinations orally.)

Text: *Performance Modeling and Design of Computer Systems*, Mor Harchol-Balter. The text is optional, but is an excellent resource. It does go well beyond the material that is covered in the course.

Course coverage:

1. Probability Introduction (Chapter 3 plus STATS 3Y03)
2. Discrete-time Markov Chains (Chapter 8 and Section 10.1)
3. Introduction to queueing (Chapter 1)
4. Operational Laws (Chapters 6 and 7)
5. The Exponential Distribution and the Poisson Process (Chapter 11)
6. Continuous-time Markov Chains (Chapter 12)
7. Simple queues (Chapter 13, Sections 14.2, 14.3, 15.2)
8. Multiserver systems
9. Networks of queues
10. Simulation
11. Reliability
12. Impact of variance on scheduling
13. Load Balancing
14. Additional Topics

The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of the student to check their McMaster email and course websites weekly during the term and to note any changes.

ACADEMIC INTEGRITY

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process.

Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: “Grade of F assigned for academic dishonesty”), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the Academic Integrity Policy, located at

<https://secretariat.mcmaster.ca/university-policies-proceduresguidelines>

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one’s own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

AUTHENTICITY / PLAGIARISM DETECTION

Some courses *may* use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. **All submitted work is subject to normal verification that standards of academic integrity have been upheld** (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

COURSES WITH AN ON-LINE ELEMENT

Some courses *may* use on-line elements (e.g. e-mail, Avenue to Learn (A2L), LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

ONLINE PROCTORING

Some courses *may* use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

CONDUCT EXPECTATIONS

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the Code of Student Rights & Responsibilities (the "Code"). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online**.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect,

disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

ACADEMIC ACCOMMODATION OF STUDENTS WITH DISABILITIES

Students with disabilities who require academic accommodation must contact Student Accessibility Services (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's Academic Accommodation of Students with Disabilities policy.

REQUESTS FOR RELIEF FOR MISSED ACADEMIC TERM WORK

McMaster Student Absence Form (MSAF): In the event of an absence for medical or other reasons, students should review and follow the Academic Regulation in the Undergraduate Calendar "Requests for Relief for Missed Academic Term Work".

ACADEMIC ACCOMMODATION FOR RELIGIOUS, INDIGENOUS OR SPIRITUAL OBSERVANCES (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the RISO policy. Students should submit their request to their Faculty Office **normally within 10 working days** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

COPYRIGHT AND RECORDING

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

EXTREME CIRCUMSTANCES

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, A2L and/or McMaster email.